

Deformable Object Simulation and Parameter Identification using Depth Observations

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Abstract—Learning bimanual manipulation of deformable materials requires models that connect camera observations, tool motion, and material response. We present **TaichiDough**, a real-to-simulation pipeline for dough observed by one calibrated RGB-D camera while two spatulas follow recorded trajectories. A metric camera–scene calibration and floor-aware reconstruction initialize a mass-preserving three-dimensional Moving Least Squares Material Point Method model. The corrected Taichi simulator replays signed-distance-field tools and differentiates a partial-view depth objective through the full particle trajectory. Material variables are optimized in scaled latent coordinates. On one recorded sequence, a five-parameter run reduced its training objective by 61.8% before interruption. We therefore interpret the estimates as configuration-dependent effective parameters rather than uniquely identified dough constants. The results establish the sensing, simulation, and optimization basis for future synthetic-data generation and robot-controlled bimanual shaping, while making the present identifiability and numerical limitations explicit.

I. INTRODUCTION

Robotic manipulation of dough and other elasto-plastic materials is challenging because each action alters both the observable geometry and the latent mechanical state of the material. This makes bimanual tool-based shaping particularly difficult: a controller must account for large deformations, complex contact interactions, rate-dependent behavior, and irreversible deformation that persists after unloading. Data-driven approaches can learn such behaviour from demonstrations or interaction data, but collecting broad physical datasets with robots is expensive. A calibrated simulator could instead generate controlled synthetic trajectories and support model-based action selection, provided that its predictions are tied to measurements of the real material.

In this work we introduce **TaichiDough**, a real-to-sim pipeline intended for future bimanual deformable object manipulation practices. A calibrated RGB-D camera records the dough while motion capture system (Mocap) provides the poses of both tools. The initial visible shape is reconstructed into a volumetric particle model, measured tool motion is replayed in a three-dimensional Material Point Method (MPM) simulator, and material parameters are adjusted to reduce disagreement between

rendered and recorded depth observations. The same forward model supports a derivative-free Young’s-modulus search and reverse-mode differentiation through complete trajectories.

The contributions are:

- a metric single-camera pipeline that combines floor-aware volumetric initialization;
- a differentiable MLS-MPM formulation for partial depth observations, including bounded optimizer coordinates for elastic, viscous, and principal-stretch parameters; and

II. RELATED WORK

Deformable object manipulation. Deformable object manipulation (DOM) spans linear, sheet-like, and volumetric objects, including ropes, cloth, and dough [1]–[3]. To fit the simulator from the available single RGB-D view and measured dual-tool trajectories, TaichiDough uses floor-aware volume reconstruction and a depth-based objective, without requiring multi-view appearance reconstruction. Physics-based approaches predict deformation for planning [4], whereas adaptive visual servoing estimates motion–deformation relationships online [5]. Learning-based methods train policies or predictive models from demonstrations and interaction data [3]. G-DOOM combines learned latent graph dynamics with model-predictive control for rope and cloth manipulation [6], showing that learned dynamics and model-based control are compatible.

Deformable simulation and differentiable physics. Simulation methods include mass–spring systems, finite-element methods, position-based dynamics, smoothed particle hydrodynamics, and the Material Point Method (MPM) [2], [3], [7]. Mass–spring models are simple but depend on network parameterization; finite-element methods discretize continuum models, while position-based dynamics enforces geometric constraints [2]. MPM accommodates large deformations without a persistently deforming mesh while retaining constitutive material models, motivating our choice for volumetric elasto-plastic simulation [7]. ChainQueen and DiffTaichi enable gradient-based optimization through simulation [8], [9], while PlasticineLab applies differentiable physics to elasto-plastic manipulation [10]. We use this capability in MLS-MPM to fit material parameters to partial depth observations.

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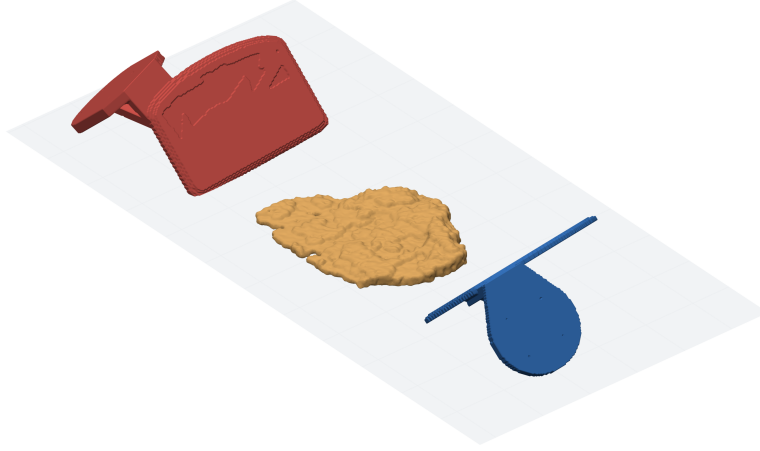


Fig. 1: Simulation setup with both watertight spatula SDFs and reconstructed dough

Completed identification pipeline

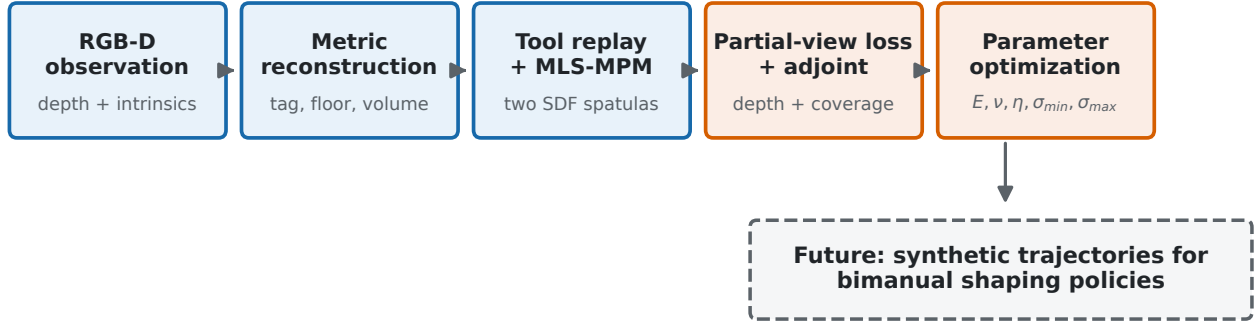


Fig. 2: TaichiDough pipeline. A single calibrated RGB-D camera and recorded poses of two spatulas provide the initial geometry, observations, and boundary motion. A corrected MLS-MPM rollout is projected into the camera and compared with measured depth. Robot execution and synthetic-data learning are intended downstream uses rather than experiments reported here.

Visual real-to-sim identification. Visual identification methods differ in sensing and geometric assumptions. DiffCloud fits simulations to point clouds, whereas Differentiable Depth uses a single depth camera and known reference geometry [11], [12]. PAC-NeRF jointly estimates geometry and physical parameters using multi-view video and differentiable MPM [13]; PhysTwin reconstructs spring-mass digital twins from interaction videos [?]. DPSI identifies elasto-plastic material and contact parameters from real interactions and incomplete point clouds [14], while EMPM combines multi-view RGB-D reconstruction with online MPM parameter refinement [15]. Within this literature, TaichiDough studies a calibrated single-view setting with floor-aware volume reconstruction, measured dual-tool trajectories, and depth-based fitting.

Robotic shaping and dough mechanics. RoboCraft and RoboCook learn predictive dynamics for goal-directed

shaping and manipulation with diverse tools [16], [17], while DoughNet predicts geometric and topological changes [18]. Complementary food-engineering studies estimate dough rheology through controlled mechanical measurements; Fabbri and Cevoli combine finite-element model inversion with extrusion measurements [19]. Dough response depends on formulation, loading, preparation, and adhesion [20], [21]. Accordingly, our fitted parameters characterize the selected constitutive and contact model over the recorded manipulation, rather than providing a complete rheological characterization.

III. METHOD

Let us formulate calibration as a trajectory-conditioned inverse problem. Let z_t denote the simulated particle state, u_t the measured poses and velocities of the two spatulas, θ the fitted material parameters, and ψ the fixed

reconstruction, contact, camera, and numerical settings:

$$\begin{aligned} z_{t+1} &= \mathcal{S}_{h,\Delta t}(z_t, u_t; \theta, \psi), \\ y_t &= \mathcal{H}_c(z_t) + \delta_t + \epsilon_t. \end{aligned} \quad (1)$$

Here $\mathcal{H}_c$ projects particles into the calibrated depth camera, while δ_t and ϵ_t denote model discrepancy and measurement noise. The estimated values are therefore effective parameters of this specified experiment and simulator, rather than universal dough constants.

A. Metric Reconstruction and Tool Replay

A single RGB-D camera records depth and colour, while the motion-capture system records the two spatula frames. Camera intrinsics and a rigid camera-to-scene transform are obtained from repeated AprilTag/mocap correspondences. The calibration also stores the table plane. All point clouds are transformed once into metric scene coordinates; no scale, translation, or rigid registration is fitted during evaluation.

The first observed dough surface is voxelised and each occupied vertical column is filled down to the calibrated table. If the reconstructed volume is V , measured dough mass is M , and N material points are sampled, then

$$\rho = M/V, \quad V_p = V/N, \quad m_p = M/N. \quad (2)$$

This initialization preserves total volume and mass when particle count is changed. Tool positions are interpolated linearly and quaternion orientations by shortest-path spherical interpolation. The spatulas are represented by signed-distance fields generated from registered meshes. Their velocity at a contact point x is

$$v_{\text{tool}}(x) = v_{\text{lin}} + \omega \times (x - c), \quad (3)$$

where c is the tool centre. The unilateral Coulomb approximation removes inward normal relative velocity and limits tangential correction using the configured friction coefficient; separating motion is left unchanged.

B. Differentiable MLS-MPM

The simulator implements three-dimensional MLS-MPM with APIC transfer [7] in Taichi [22]. Each particle stores position x_p , velocity v_p , deformation gradient F_p , affine velocity matrix C_p , and optional plastic-volume state. The deformation update is given with:

$$F_p^{n+1} = (I + \Delta t C_p) F_p^n. \quad (4)$$

Young's modulus E and Poisson ratio ν define:

$$\mu = \frac{E}{2(1+\nu)}, \quad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}. \quad (5)$$

With polar factor R , $J = \det(F_p)$, and viscosity η , the implemented stress contribution is:

$$P = 2\mu(F_p - R)F_p^T + \lambda J(J-1)I + \eta(C_p + C_p^T). \quad (6)$$

The final term is an additive rate stress, not a multi-time-scale rheological model. For quadratic B-spline weight

w_{pi} , particle-to-node offset d_{pi} , and grid spacing Δx , particle-to-grid transfer accumulates

$$m_i += w_{pi} m_p, \quad (7)$$

$$m_i v_i += w_{pi} [m_p v_p + (-4\Delta t V_p \Delta x^{-2} P + m_p C_p) d_{pi}]. \quad (8)$$

The corrected grid-to-particle update reconstructs the affine velocity using $4\Delta x^{-2} d_{pi}$. Optional plasticity decomposes $F_p = U\Sigma V^T$ and clamps each principal stretch to $[\sigma_{\min}, \sigma_{\max}]$ before reconstruction. These bounds are dimensionless stretch limits, not yield stresses.

C. Partial-View Objective and Optimization

At each selected frame, a differentiable camera-space particle renderer uses compact splats and soft visibility to produce depth, coverage, and visible particle coordinates in the calibrated camera. It produces depth, coverage, and visible particle coordinates in the calibrated camera. After normalizing the configured weights by their sum, the per-frame objective is

$$\mathcal{L}_t = 0.5\ell_d + 0.25\ell_c + 0.25\ell_q, \quad (9)$$

where ℓ_d compares simulated and observed depth change relative to the initial frame, ℓ_c penalizes missing positive observed foreground, and ℓ_q is the directed distance from observed points to visible predicted particles. Unknown pixels are not treated as empty space.

Gradients of the trajectory loss with respect to the material parameters are computed using checkpointed reverse-mode differentiation. Simulator states are stored at selected time steps, while intermediate states are recomputed during the backward pass to limit memory use. The physical parameters are represented by bounded, scaled optimization variables, with a logarithmic transformation for Young's modulus because of its broad admissible range. A bound-constrained gradient optimizer with backtracking then updates the parameters while rejecting invalid or non-decreasing steps.

D. Data Collection

Data were collected using two 3D-printed spatulas. The experimental setup consisted of an Intel RealSense D435 camera and an OptiTrack motion-capture system. OptiTrack markers were attached to the spatulas to track their position and orientation during manipulation.

The transformation between the spatula mesh and the corresponding OptiTrack marker frame was determined through a calibration procedure using UR5e and Kinova Gen3 robotic manipulators. During calibration, a set of robot waypoints was recorded together with the corresponding marker poses measured by the OptiTrack system. These paired measurements were then used to estimate the transformation between the spatula mesh frame and the OptiTrack marker frame.

IV. RESULTS AND DISCUSSION

Figs. 4 and 5 show the camera-matched sequence and differentiable optimization history.



Fig. 3: Operator shaping the dough into a ball using two spatulas during data collection. OptiTrack markers attached to the dough are visible.

A. Static and Differentiable Fitting

Differentiable run jointly adjusts E , ν , viscosity η , and the two principal-stretch bounds. Seventeen objective evaluations and fifteen accepted updates reduce the partial-view training objective from 0.42455 to 0.16229. The best saved state is

$$\begin{aligned} E &= 8.803 \text{ kPa}, & \nu &= 0.48708, & \eta &= 30.37 \text{ Pa}\cdot\text{s}, \\ \sigma_{\min} &= 0.86262, & \sigma_{\max} &= 1.06714. \end{aligned} \quad (10)$$

Figure 4 compares frames from the recorded experiment (top row) with the corresponding MPM particle renderings (bottom row). The simulation closely reproduces the observed dough deformation during the first second, but the agreement deteriorates by ($t=2$,s), after which the simulated dough exhibits little further deformation.

B. Limitations for Single-View Identification

A single depth view leaves hidden volume and occluded contact unmeasured, and the evaluator does not render the spatulas into the simulated depth image. Material estimates are consequently conditional on reconstructed volume, mass/density, floor pose, tool registration, contact law and friction, numerical resolution, and loss smoothing. A separate diagnostic also finds an observed table plane about 47.6mm above the configured floor with approximately 5° tilt, which can be absorbed incorrectly by material or contact parameters. Near-incompressible ν , elasticity, viscosity, and stretch clipping may compensate for one another under a single monotonic trajectory. The present evidence therefore supports an implemented observation-conditioned fitting pipeline and measurable prediction improvement, but not camera-only rheometry or universal dough constants.

C. Limitations of no force measurements

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V. CONCLUSION

TaichiDough connects a calibrated RGB-D recording and measured two-spatula motion to a mass-preserving MLS-MPM model, partial-view differentiable objective, bounded parameter optimizer, and independent visible-geometry evaluation. Existing results show millimetric centroid/depth alignment with 0.831 footprint IoU, substantial reduction of the differentiable training objective, and better temporal-continuation prediction than frozen geometry. They also show why objective reduction alone is insufficient for physical identification: the derivative-free modulus search is flat, one differentiable result is interrupted, another reaches a parameter bound, and full CUDA replay is not exactly reproducible.

The current values are therefore effective parameters of a stated camera, reconstruction, contact law, constitutive model, and discretization. Future work will use robot-controlled spatulas to collect deliberately informative motions; repeat measurements across dough preparations; model tool occlusion; qualify full-horizon gradients; and compare matched differentiable and derivative-free optimizers. Once prediction is validated on independent manipulations, the simulator can support synthetic pre-training for inverse models and bimanual dough-shaping policies.

REFERENCES

- [1] J. Sanchez, J.-A. Corrales, B.-C. Bouzgarrou, and Y. Mezouar, "Robotic manipulation and sensing of deformable objects in domestic and industrial applications: A survey," *The International Journal of Robotics Research*, vol. 37, no. 7, pp. 688–716, 2018.
- [2] V. E. Arriola-Rios, P. Guler, F. Ficuciello, D. Kragic, B. Siciliano, and J. L. Wyatt, "Modeling of deformable objects for robotic manipulation: A tutorial and review," *Frontiers in Robotics and AI*, vol. 7, p. 82, 2020.
- [3] H. Yin, A. Varava, and D. Kragic, "Modeling, learning, perception, and control methods for deformable object manipulation," *Science Robotics*, vol. 6, no. 54, p. eabd8803, 2021.
- [4] P. Jiménez, "Survey on model-based manipulation planning of deformable objects," *Robotics and Computer-Integrated Manufacturing*, vol. 28, no. 2, pp. 154–163, 2012.
- [5] D. Navarro-Alarcon, H. M. Yip, Z. Wang, Y.-H. Liu, F. Zhong, T. Zhang, and P. Li, "Automatic 3-d manipulation of soft objects by robotic arms with an adaptive deformation model," *IEEE Transactions on Robotics*, vol. 32, no. 2, pp. 429–441, 2016.
- [6] X. Ma, D. Hsu, and W. S. Lee, "Learning latent graph dynamics for visual manipulation of deformable objects," in *2022 International Conference on Robotics and Automation (ICRA)*, pp. 8266–8273, IEEE, 2022.
- [7] C. Jiang, C. Schroeder, J. Teran, A. Stomakhin, and A. Selle, "The material point method for simulating continuum materials." *ACM SIGGRAPH 2016 Courses*, 2016.
- [8] Y. Hu, J. Liu, A. Spielberg, J. B. Tenenbaum, W. T. Freeman, J. Wu, D. Rus, and W. Matusik, "Chainqueen: A real-time differentiable physical simulator for soft robotics," in *IEEE International Conference on Robotics and Automation*, 2019.
- [9] Y. Hu, L. Anderson, T.-M. Li, Q. Sun, N. Carr, J. Ragan-Kelley, and F. Durand, "DiffTaichi: Differentiable programming for physical simulation," in *International Conference on Learning Representations*, 2020.
- [10] Z. Huang, Y. Hu, T. Du, S. Zhou, H. Su, J. B. Tenenbaum, and C. Gan, "Plasticinlab: A soft-body manipulation benchmark with differentiable physics," in *International Conference on Learning Representations*, 2021.

Camera-matched Episode 18 observation and simulation

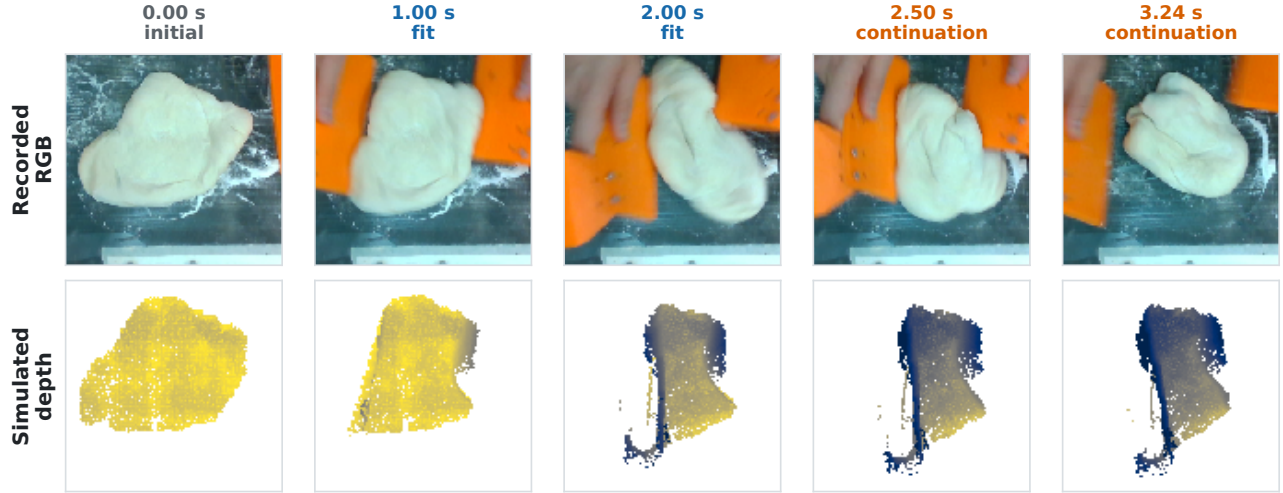


Fig. 4: Exact-timestamp Episode 18 camera views and simulation outputs. The top row is contextual RGB extracted from the original Rosbag; the bottom row is saved dough-only simulated optical depth rendered with the same calibrated camera and display crop. The simulation does not render spatula occlusion. Frames after 2.00s are temporal continuation of the same motion, not independent generalization.

- [11] P. Sundaresan, R. Antonova, and J. Bohg, “Diffcloud: Real-to-sim from point clouds with differentiable simulation and rendering of deformable objects,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2022.
- [12] K. Arnavaz, M. K. Nielsen, P. G. Kry, M. Macklin, and K. Erleben, “Differentiable depth for real2sim calibration of soft body simulations,” *Computer Graphics Forum*, 2023.
- [13] X. Li, Y.-L. Qiao, P. Y. Chen, K. M. Jatavallabhula, M. Lin, C. Jiang, and C. Gan, “PAC-NeRF: Physics augmented continuum neural radiance fields for geometry-agnostic system identification,” in *The Eleventh International Conference on Learning Representations (ICLR)*, 2023.
- [14] X. Yang, Z. Ji, and Y.-K. Lai, “Differentiable physics-based system identification for robotic manipulation of elastoplastic materials,” *The International Journal of Robotics Research*, 2025.
- [15] Y. Chen, Y. Hu, L. Sun, T. Kusunur, L. Herlant, and C. Jiang, “EMPM: Embodied MPM for modeling and simulation of deformable objects,” *IEEE Robotics and Automation Letters*, 2026.
- [16] H. Shi, H. Xu, Z. Huang, Y. Li, and J. Wu, “Robocraft: Learning to see, simulate, and shape elasto-plastic objects with graph networks,” in *Robotics: Science and Systems*, 2022.
- [17] H. Shi, H. Xu, S. Clarke, Y. Li, and J. Wu, “Robocook: Long-horizon elasto-plastic object manipulation with diverse tools,” in *Conference on Robot Learning*, 2023.
- [18] D. Bauer, Z. Xu, and S. Song, “Doughnet: A visual predictive model for topological manipulation of deformable objects,” in *European Conference on Computer Vision*, 2024.
- [19] A. Fabbri and C. Cevoli, “Estimation of semolina dough rheological parameters by inversion of a finite elements model,” *Journal of Agricultural Engineering*, vol. 46, no. 3, pp. 95–99, 2015.
- [20] G. Yazar, “Wheat flour quality assessment by fundamental non-linear rheological methods: A critical review,” *Foods*, vol. 12, no. 18, 2023.
- [21] D. Ghorbel and B. Launay, “An investigation into the nature of wheat flour dough adhesive behaviour,” *Food Research International*, 2014.
- [22] Y. Hu, T.-M. Li, L. Anderson, J. Ragan-Kelley, and F. Durand, “Taichi: A language for high-performance computation on spatially sparse data structures,” *ACM Transactions on Graphics*, vol. 38, no. 6, 2019.

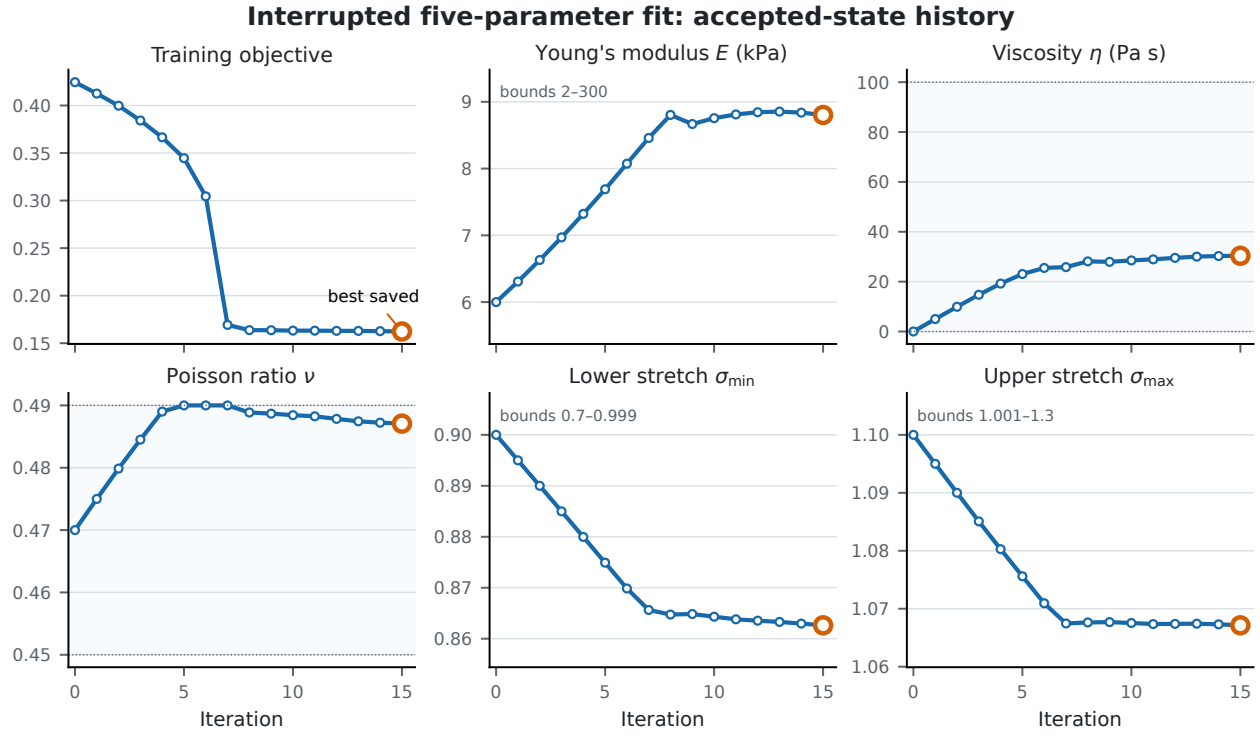


Fig. 5: Accepted objective and parameter history for the five-variable differentiable run. The orange ring marks the best saved state.